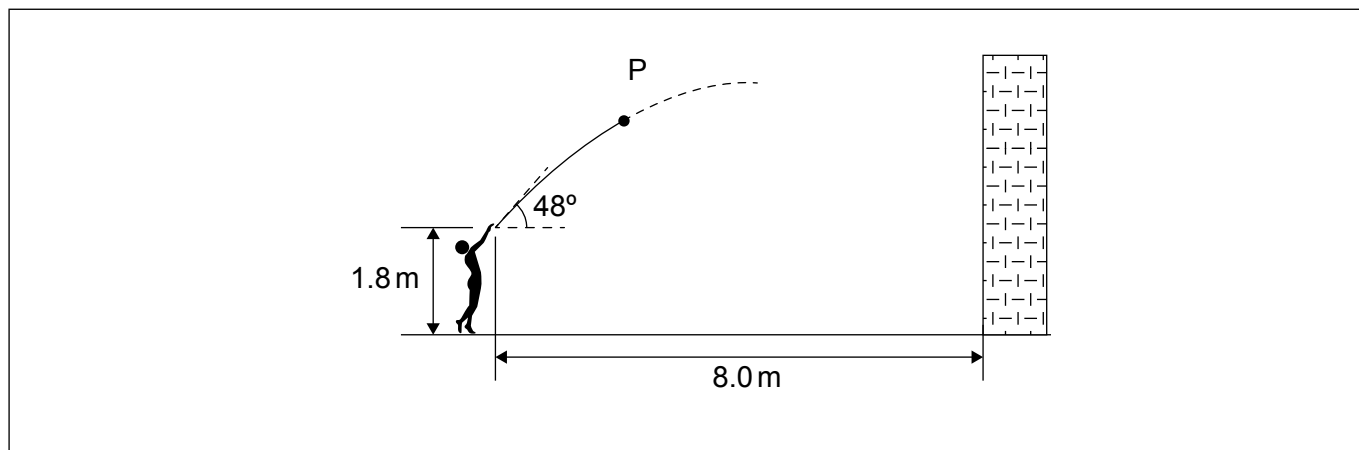


13. A student throws a ball towards a wall. The ball is released from a point 1.8 m above the ground and 8.0 m from the wall. The initial velocity of the ball makes an angle of 48° with the horizontal. Air resistance is negligible.

The diagram shows the initial path of the ball. P is a point on the path.



- (a) Draw, on the diagram, an arrow to show
- (i) the velocity of the ball at P. Label this arrow v . [1]
 - (ii) the acceleration of the ball at P. Label this arrow a . [1]

The ball takes 1.3 s to reach the wall.

- (b) (i) Show that the initial speed of the ball is about 9 m s^{-1} . [2]

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- (ii) Determine the height above the ground at which the ball hits the wall. [3]

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